Tensors and Exterior Algebra

Three times already this book has built an object out of one construction and then walked past the construction without naming it. Chapter 7 characterized the determinant as the **unique** alternating n-linear form sending the standard basis to 1 (Theorem 7.4.2), and got a number out of a function of n vectors. Chapter 8 defined the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ by writing $\mathbf{A}$ with its entries replaced by blocks, borrowed the symbol $\otimes$, and admitted that the symbol was borrowed from something it had not yet met. Chapter 14 defined a Clifford algebra by saying what maps out of it must look like, proved that at most one such algebra exists, and then could not produce one: the construction needed an algebra of "formal sums of strings of vectors", and no such algebra had been built. The missing object is the same in all three places. It is the **tensor product**, and this chapter builds it.

The problem it solves is stated in one line. A bilinear map $\beta\colon V \times W \to Z$ is not a linear map, so nothing from Chapters 2 to 5 applies to it: it has no matrix, no kernel, no rank, and $\beta(2v, 2w)$ is $4\beta(v, w)$ rather than $2\beta(v, w)$. Multilinearity is everywhere in the book — the determinant, the dot product, matrix multiplication, every bilinear form of Chapter 14 — and each time we have had to develop it by hand, because the general theory of linear maps was unavailable. The tensor product ends that. It manufactures a single space $V \otimes W$ and a single bilinear map $(v, w) \mapsto v \otimes w$ into it, through which **every** bilinear map out of $V \times W$ factors, uniquely, as a linear map. Bilinear maps on $V \times W$ become linear maps on $V \otimes W$. From there the whole chapter is bookkeeping: repeat the construction k times to get the tensor algebra, then divide it by the relation $v \otimes w = w \otimes v$ to get symmetric powers, or by the relation $v \otimes v = 0$ to get exterior powers, where the determinant reappears as an induced map and the Clifford algebra finally gets built.

The method of this chapter is new, and §02 is where it is learned. Until now every object in the book was defined by a formula or by a construction: the determinant by the Leibniz sum, the quotient by cosets, the Jordan form by an algorithm. From here on we define objects by what maps out of them look like — a **universal property** — and the order of business inverts. First we say what the object must do; then we prove that any two objects doing it are isomorphic by a *unique* isomorphism, so that "the" object is legitimate; and only then do we build one, to know that the definition is not empty. Chapter 14 §12 already ran this argument once, for Clifford algebras (Proposition 14.12.1), without saying that it was a pattern. It is a pattern. Sections 02, 06, 07, 08, and 10 each execute it in exactly the same three moves, on five different objects. Theorem 15.2.1 in §02 is the template, written out at full length; the four later repetitions are compressed, and they are compressed on purpose, so that the reader who has absorbed the template recognizes it rather than rereads it. It is worth learning here because it is the only way to define these objects that does not depend on a choice of basis, and because a proof from a universal property is usually three lines long where a proof from a construction is a page.

The field is a general F throughout. Where a statement genuinely needs $\operatorname{char} F = 0$ or $\operatorname{char} F \neq 2$ — and some do, in §§07, 08, and 10 — the hypothesis is stated and its job is named, as Chapter 14 did. Everything else holds over every field.

What you need. From Chapter 2, the space of linear maps (Definition 2.4.1, Theorem 2.4.1) and rank with Rank–Nullity (Definition 2.3.2, Theorem 2.3.1). From Chapter 4, quotient spaces and their universal property, which is the engine of every construction here (Definition 4.4.3, Lemma 4.4.3, Theorem 4.4.4, Theorem 4.4.10, Theorem 4.4.5, Theorem 4.4.7), the space of linear maps seen as matrices (Theorem 4.1.4), the matrix of a map (Definition 4.1.1, Theorem 4.1.1), products of spaces (Definition 4.4.1, Theorem 4.4.1), and the trace (Definition 4.5.2,

Theorem 4.2.5). From Chapter 1, bases in any dimension and the freedom to prescribe a linear map on one (Definition 1.5.1, Theorem 1.6.4, Theorem 1.8.2, Theorem 2.1.6, Theorem 1.2.1). From Chapter 5, duality, used constantly: Definition 5.1.2, Definition 5.1.1, Theorem 5.1.3, Theorem 5.1.5, Definition 5.4.1, Theorem 5.4.2, Theorem 5.4.6. From Chapter 7, multilinearity and alternation, which §01 extends rather than redefines (Definition 7.2.1, Definition 7.2.2, Definition 7.2.3, Theorem 7.2.1, Theorem 7.2.2), and the determinant with its Leibniz formula and multiplicativity (Definition 7.4.1, Theorem 7.4.3, Theorem 7.4.2, Theorem 7.5.4, Theorem 7.4.5, Theorem 7.6.1, Theorem 7.7.1). From Chapter 8, the Kronecker product, whose two open promises §03 pays (Definition 8.4.1, Theorem 8.4.1, Theorem 8.4.3, Definition 8.4.2, Theorem 8.4.4). From Chapter 13, the singular value decomposition, which §11 turns out to be reading in another language (Theorem 13.9.1), and, for the same section, from Chapter 11 the orthogonal apparatus that goes with it: the inner product and Pythagoras, Gram–Schmidt and orthonormal coordinates, and the orthogonal decomposition (Definition 11.1.1, Theorem 11.1.3, Theorem 11.2.2, Theorem 11.2.1, Theorem 11.3.2). From Chapter 14, algebras, forms, and the Clifford algebra whose existence §10 supplies (Definition 14.12.1, Definition 14.1.1, Definition 14.3.2, Definition 14.3.1, Definition 14.12.2, Proposition 14.12.1, Proposition 14.12.3, Theorem 14.12.4). From Chapter 6, the polynomial ring, which §07 identifies with a symmetric algebra (Definition 0.8.2).

15.-1.1 Roadmap

- **Multilinear maps.** Chapter 7's multilinear forms, extended from $V^k \to F$ to $V_1 \times \cdots \times V_k \to W$; the space they form, its dimension $\dim W \cdot \prod_i \dim V_i$, and the problem that they are not linear maps.
- **The tensor product.** The universal property, uniqueness up to unique isomorphism, the construction as a free space modulo the bilinearity relations, the basis $(v_i \otimes w_j)$, and three warnings about what a tensor is not.
- **Tensor products of maps.** $S \otimes T$ built from the universal property; its matrix is exactly the Kronecker product of Chapter 8, which settles two debts; and the identification $V^* \otimes W \cong \mathcal{L}(V, W)$ that everything later uses.
- **Contraction.** Evaluation $V^* \otimes V \to F$, extended to mixed tensors, and the discovery that under the previous identification it is the trace — the basis-free definition of trace the book has wanted since Chapter 4.
- **Index notation.** Components, upper and lower indices, the summation convention, the transformation law derived rather than asserted, and the older definition of a tensor proved equivalent to ours.
- **The tensor algebra.** $T(V) = \bigoplus_k V^{\otimes k}$: graded, associative, noncommutative, infinite-dimensional, and free. Its universal property is what the next three sections quotient.
- **Symmetric powers.** $\operatorname{Sym}^k V$ by killing $v \otimes w - w \otimes v$; a basis indexed by nondecreasing tuples; the symmetric algebra is the polynomial ring of Chapter 6; and what goes wrong in characteristic p.
- **Exterior powers.** $\Lambda^k V$ by killing $v \otimes v$ — the characteristic-free choice, exactly Chapter 14's distinction between alternating and skew. A basis indexed by increasing tuples, $\dim \Lambda^k V = \binom{n}{k}$, and the test that $v_1 \wedge \cdots \wedge v_k \neq 0$ exactly when the list is independent.
- **The determinant, again.** $\Lambda^n V$ is a line, and $\Lambda^n T$ is multiplication by $\det T$. Multiplicativity becomes a one-line consequence, Laplace expansion falls out of Λ^k, and the coordinates of a wedge are the minors.
- **Grassmann and Clifford.** The exterior algebra ΛV, of dimension 2^n; elements of $\Lambda^2 V$ that are not decomposable; and $\operatorname{Cl}(q) = T(V)/\langle v \otimes v - q(v)1 \rangle$, which constructs the Clifford algebra of Chapter 14 and proves its dimension is 2^n.
- **Tensors in quantum mechanics.** Composite systems as tensor products, entangle-

ment as failure of simplicity, and the Schmidt decomposition, which is the singular value decomposition of Chapter 13 wearing different notation.

15.-1.2 Named moves

- **Define by universal property, then build a model.** State what maps out of the object must look like. Prove that any two objects with that property are isomorphic by a unique isomorphism, using nothing but the property itself. Then construct one object, so that the definition has content. The uniqueness proof is always the same four lines: each object's property produces a map to the other, each composite is a self-map fixing the structure map, and the uniqueness clause says the identity is the only such self-map.
- **A quotient kills exactly the relations you want.** To impose relations on a space, take the free object, take the subspace spanned by the relations written as "left side minus right side", and divide. A map respecting the relations kills that subspace, so it factors through the quotient. This chapter does it three times: tensor, symmetric, exterior.
- **To define a map out of a quotient, define it upstairs first.** Never define a map on cosets, or on simple tensors, by a formula and hope it is well defined. Define it on the free object, where a map may be prescribed arbitrarily on a basis, check that it kills the relations, and let the universal property hand you the induced map.
- **Count a basis by counting index tuples.** Every dimension formula in this chapter comes from counting tuples: all k-tuples for $V^{\otimes k}$, giving n^k; nondecreasing ones for $\operatorname{Sym}^k V$, giving $\binom{n+k-1}{k}$; strictly increasing ones for $\Lambda^k V$, giving $\binom{n}{k}$. Spanning is a rewriting argument, independence is a supply of linear functionals, and the count is combinatorics.